

Agenda

Fundamental Terms and Concepts
Installing and setting up python
Literals, Variables, and Numeral Systems
Input and Output in Python (Console I/O)
Operators and Expressions

Timings:

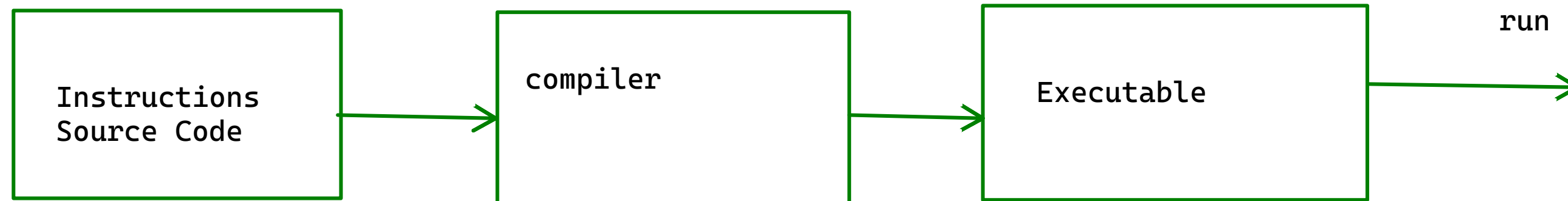
9.30AM to 11.00
15Min Break
11.15 to 1.00
1Hr Lunch break
2.00 to 3.45
15Min Break
4.00 to 5.30

Instructions in High Level language → 1/0 (Binary)

Two Ways

1. Compilation
2. Interpretation

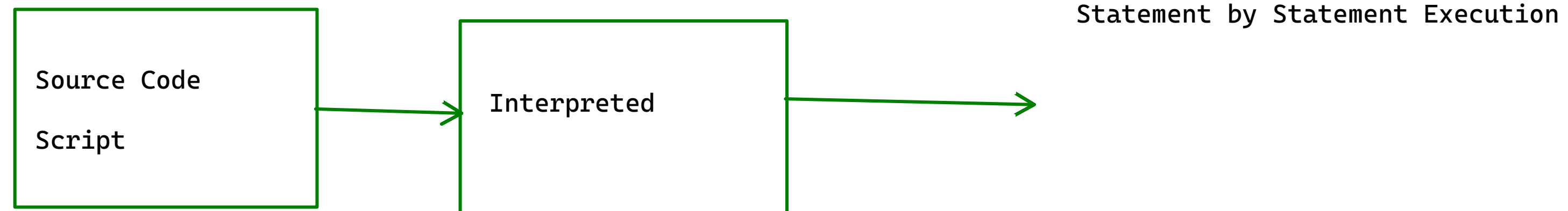
Compilation



Examples: C, C++

Linking,
Syntax (Rules for giving instructions) Check
Convert to Binary

Interpretation



Examples: Python, JavaScript

Pros & Cons

Compilation:

- 1. Process takes time for compilation, Execution is faster
- 2. Any User need not have additional software to run
- 3. More granular control
- 4. much secure than interpreted languages

Interpretation:

- 1. Execution of each command might take time compared to compilations
- 2. No hardware level granular control
- 3. source code (script) is shared to user
- 4. Needs an additional software (Interpreter)

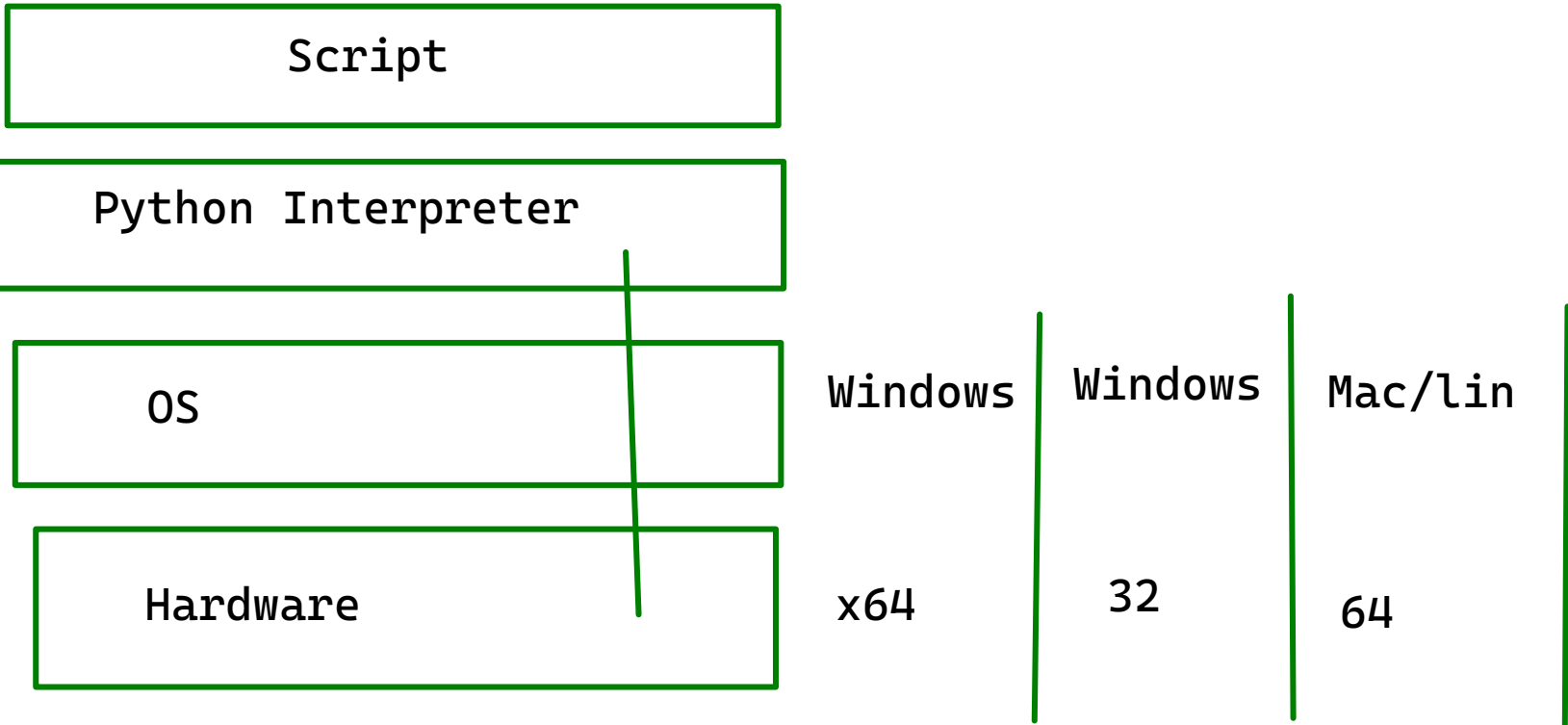
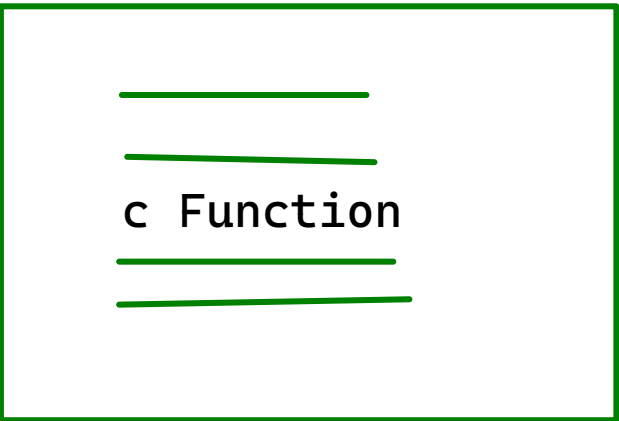
Interpreter stack in Python

Python is a
Interpreted Language
General Purpose Language
Dynamically typed language
Open Source Language
WORA – Write Once Run Any Where
Script is Independent of Arcitecture

LGPL
GPL
MIT

Types of Interpreters

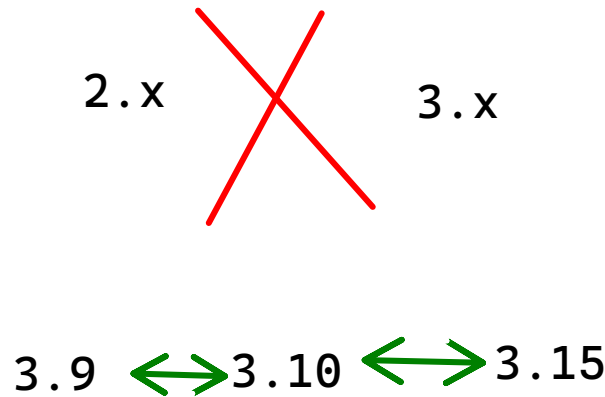
- 1. Cython
- 2. Jython
- 3. RPython
- 4. Pypi



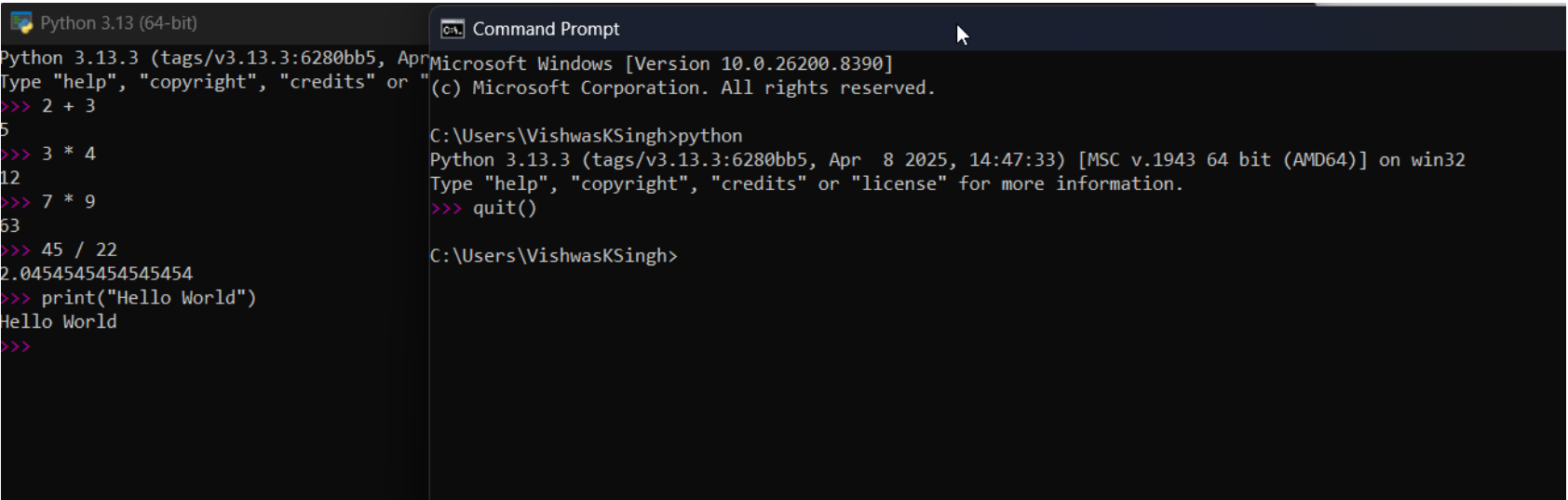
Versions of Python

2.x

3.x (Latest)



Not backward Compatible



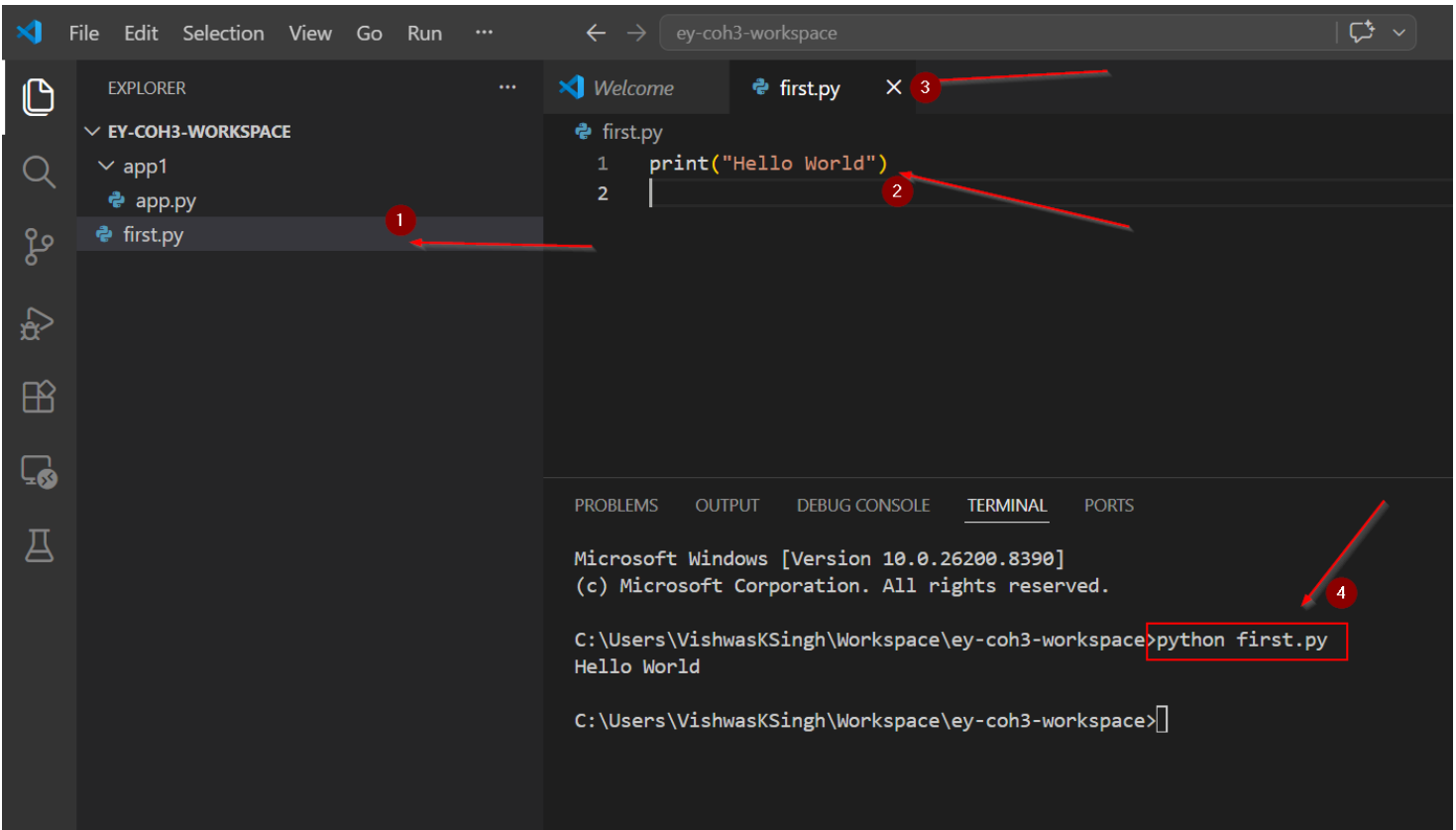
Things to Run & Develop a Python program

1. Python interpreter
2. Text Editor

Interactive Mode

Ways of Interacting with python interpreter

1. Interactive mode
2. Script Mode
3. Notebook mode



Script Mode

Problem:

Analyst, Some data in webpage, Csv, Excel File, Combine those & generate a report

Sub Problems

1. Data is multiple sources - to be read

sub problem:

a. Excel File Source - Where is the File, Read the File, Send the data to next step - Function For these Steps

b. Website

2. Process it

3. Generate a report

Python Language

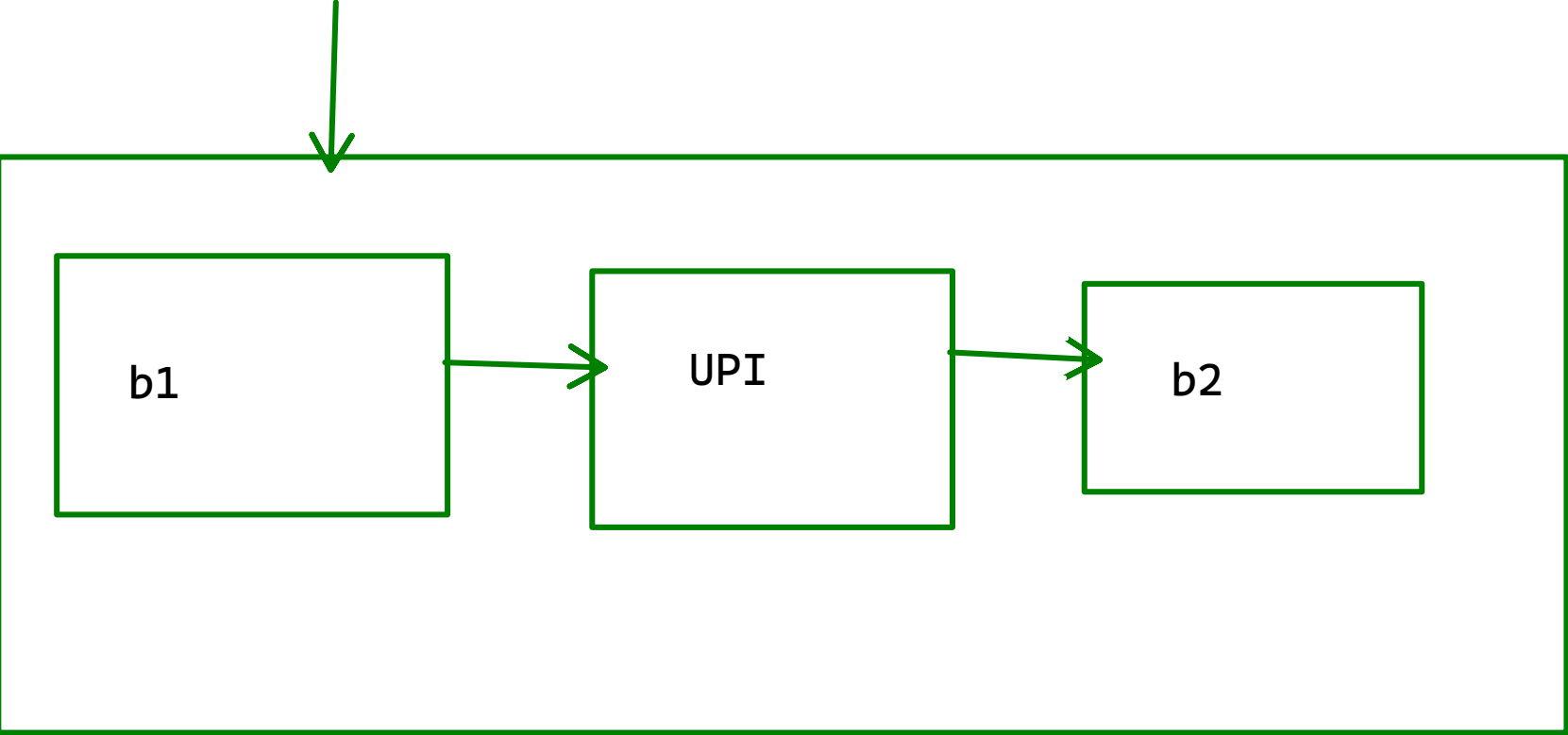
- 1. Why was it invented - Base Problem way
- 2. How to store data temporarily
- 3. How to do Conditions
- 4. How to Repetition

1. Functional Approach

but is object based

Withdraw()
Deposit()

How much amout from which bank to bank



OOPS - Object oriented Approach - Java

Banking - 100s customers

Identify the nouns - Customer, account, bank branch ...
Classes - Customer, account,

Customer	account
Cust name	acc_id
id	cust_id
acc_balance	

Can Deposit()
Can withdraw()

Create Objects -

Problem

You are saving an amount in savings bank account
gives - 5% interest per annum

Input:	Output:	Logic:
principal: 300000 rate_of_interest: 5% time_in_months: 36	Interest_amount	func(p,t,r) simple_interest = principal * rate_of_interest * time_in_months return simple

How to Save data temporarily:

Variables → identifier(Label) + Value

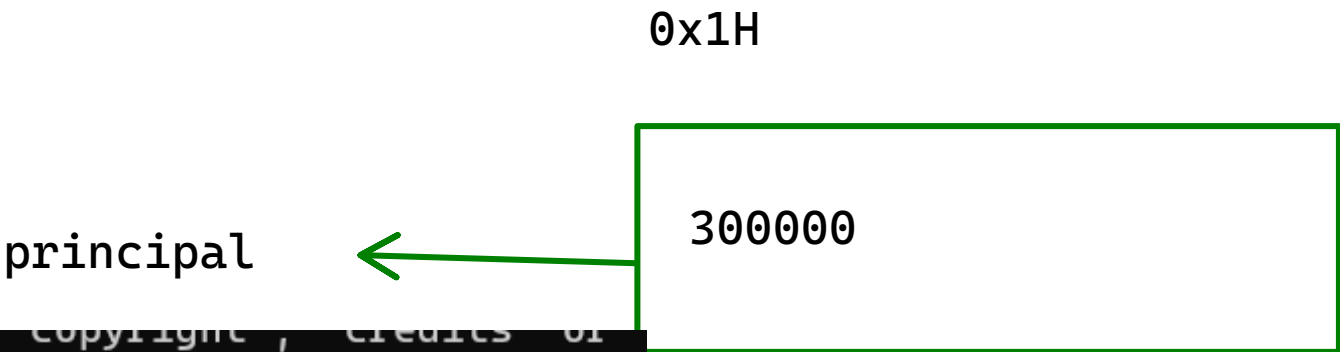
principal = 300000

Rules of a Variable:

- 1. It should always start with letter or _
- 2. It can contain number, _
- 3. It cannot be a key word
- 4. Case Sensitive

3 Types of Values: Data Type

- 1. Number (int, float)
- 2. String
- 3. Boolean



```
type help , copyright , credits of  
>>> principal = 300000  
>>> type(principal)  
<class 'int'>  
>>> rate_of_interest = 0.07  
>>> type(rate_of_interest)  
<class 'float'>  
>>> nme = "vishwas"  
>>> type(nme)  
<class 'str'>  
>>> isValid = True  
>>> type(isValid)  
<class 'bool'>  
>>>
```

```
>>> cp = complex(2,3)
>>> cp
(2+3j)
>>> type(cp)
<class 'complex'>
>>> hexnum = 0x1A
>>> type(hexnum)
<class 'int'>
>>> hexnum
26
>>> binnum = 0b1010
>>> binnum
10
>>> type(binnum)
<class 'int'>
>>> k = 152364575937264845756383628274628263747562826473826384632628374638263846382638463836482628473827393748478273645\
85758459374
>>> k
152364575937264845756383628274628263747562826473826384632628374638263846382638463836482628473827393748478273645857584593
74
>>> k1 = 6.3e10
>>> k1
63000000000.0
>>> type(k1)
<class 'float'>
>>> p = 20_00_000
>>> p
2000000
>>> |
```

Conventions of Naming:

1. Always have meaningful names
2. variables with multiple words follow snake_case

Operators in Python

Any number which is other than 0 is always true

Zero is always false

The input taken from a user is always a string, developer has to convert to right type

Print Function

1. Can take any number of values
2. By default it inserts a new line
3. It inserts a space as a separator by default , it can be overridden by sep="Value"

format specifiers: %d, %f, %s etc ...

Once a character # appears in a statement – the sequence following it is considered a comment

Comments are not processed

Comments are for developers or users who use your code

Standard to maintain readability PEP8 Style Guide

Linting Process – pylint(pep8), pylance, flake8